

NUMPY & PANDAS MINI PROJECTS

Project 1: Student Performance Analysis

Analyze the marks of students in different subjects using NumPy and convert the final results into a Pandas DataFrame.

Given Dataset

```
marks = np.array([
    [85, 80, 90],
    [70, 75, 65],
    [92, 88, 95],
    [60, 72, 68],
    [78, 82, 80]
])
```

Columns: Python | SQL | Machine Learning

Questions / Tasks

1. How will you calculate the total marks obtained by each student?
2. How will you calculate the average marks of each student?
3. How will you calculate the average marks in each subject?
4. How will you find the highest score in each subject?
5. How will you find the lowest score in each subject?
6. How will you find students whose average marks are above 80?
7. How will you use np.where() to assign Pass or Fail status?
8. How will you find the index of the highest-performing student?
9. How will you calculate the standard deviation for each subject?
10. How will you convert the final results into a Pandas DataFrame?

Project 2: Employee Salary Analysis

Analyze employee salary, experience and performance data.

Given Dataset

```
salary = np.array([
    [25000, 2, 80],
    [45000, 5, 90],
    [30000, 3, 75],
    [60000, 8, 95],
    [35000, 4, 85]
])
```

Columns: Salary | Experience (Years) | Performance Score

Questions / Tasks

1. How will you calculate the average salary of all employees?
2. How will you find the highest salary?
3. How will you find the lowest salary?
4. How will you calculate the average experience?
5. How will you find employees whose salary is greater than ₹40,000?
6. How will you find employees whose performance score is greater than 80?
7. How will you find the employee with the highest performance score?
8. How will you calculate the standard deviation of salaries?
9. How will you classify employees as High Salary or Low Salary using np.where()?
10. How will you convert the results into a Pandas DataFrame?

Project 3: Sales Performance Analysis

Analyze monthly sales performance of five salespersons.

Given Dataset

```
sales = np.array([
    [12000, 15000, 18000],
    [10000, 14000, 16000],
    [18000, 20000, 22000],
    [9000, 12000, 15000],
    [15000, 17000, 19000]
])
```

Columns: January | February | March

Questions / Tasks

1. How will you calculate the total sales of each salesperson?
2. How will you calculate the average monthly sales for each salesperson?
3. How will you find the highest sales achieved in each month?
4. How will you find the lowest sales achieved in each month?
5. How will you find the salesperson with the highest total sales?
6. How will you find salespersons whose average sales are above ₹15,000?
7. How will you calculate the total company sales for each month?
8. How will you calculate the standard deviation of monthly sales?
9. How will you classify salespersons as Excellent, Good or Needs Improvement using np.where()?
10. How will you convert the final results into a Pandas DataFrame?

Project 4: Student Attendance Analysis

Analyze attendance percentages of students across four technical subjects.

Given Dataset

```
attendance = np.array([
    [90, 85, 95, 88],
    [75, 80, 70, 78],
    [95, 92, 98, 96],
    [65, 70, 72, 68],
    [85, 88, 90, 87]
])
```

Columns: Python | Java | SQL | ML

Questions / Tasks

1. How will you calculate the average attendance of each student?
2. How will you calculate the average attendance of each subject?
3. How will you find the highest attendance in each subject?
4. How will you find the lowest attendance in each subject?
5. How will you find students having average attendance above 80%?
6. How will you find the student with the highest average attendance?
7. How will you calculate the standard deviation of attendance?
8. How will you use np.where() to classify students as Eligible or Not Eligible based on 75% attendance?
9. How will you convert the data into a Pandas DataFrame?
10. How will you add a Status column to the DataFrame?

Project 5: Product Sales Analysis

Analyze monthly sales of different products. This project combines NumPy calculations with Pandas DataFrame operations.

Given Dataset

```
sales = np.array([
    [100, 120, 150],
    [80, 100, 130],
    [150, 160, 180],
    [70, 90, 110],
    [120, 140, 160]
])
```

Columns: January | February | March

Questions / Tasks

1. How will you calculate total sales for every product?
2. How will you calculate average monthly sales for every product?
3. How will you find the best-selling product?
4. How will you find the worst-selling product?
5. How will you find the highest sales in each month?
6. How will you find the lowest sales in each month?
7. How will you find products whose average sales are greater than 120?
8. How will you calculate total sales for each month?
9. How will you calculate the standard deviation of monthly sales?
10. How will you create High/Low product classifications using np.where()?
11. How will you create a Pandas DataFrame containing Product, Total Sales and Average Sales?
12. How will you sort the DataFrame according to Total Sales?

Project 6: Weather Data Analysis

Analyze temperature data for four cities over five days.

Given Dataset

```
temperature = np.array([
    [32, 34, 35, 33, 31],
    [28, 30, 31, 29, 27],
    [35, 36, 38, 37, 34],
    [25, 27, 29, 28, 26]
])
```

Columns: Monday | Tuesday | Wednesday | Thursday | Friday

Questions / Tasks

1. How will you calculate the average temperature of each city?
2. How will you find the maximum temperature recorded by each city?
3. How will you find the minimum temperature recorded by each city?
4. How will you identify the hottest day?
5. How will you identify the coolest day?
6. How will you find the city with the highest average temperature?
7. How will you find cities whose average temperature is greater than 30°C?
8. How will you calculate the standard deviation?
9. How will you classify each temperature as Hot or Normal using np.where()?
10. How will you convert the data into a Pandas DataFrame?

Project 7: Bank Transaction Analysis

Analyze customer deposits, withdrawals and investments.

Given Dataset

```
transactions = np.array([
    [5000, 2000, 3000],
    [8000, 1500, 4000],
    [3000, 1000, 2500],
    [10000, 3000, 5000],
    [7000, 2500, 3500]
])
```

Columns: Deposit | Withdrawal | Investment

Questions / Tasks

1. How will you calculate the total deposit amount?
2. How will you calculate the total withdrawal amount?
3. How will you calculate the total investment amount?
4. How will you calculate the net balance using Deposit - Withdrawal?
5. How will you find the customer with the highest deposit?
6. How will you find the customer with the highest withdrawal?
7. How will you find customers whose net balance is greater than ₹4,000?
8. How will you calculate the average deposit?
9. How will you calculate the standard deviation of deposits?
10. How will you classify customers as High, Medium or Low based on net balance?
11. How will you convert the final data into a Pandas DataFrame?

Project 8: E-Commerce Customer Analysis

Analyze customer age, number of orders and total spending. Use NumPy for numerical analysis and Pandas for final tabular analysis.

Given Dataset

```
customers = np.array([
    [25, 5, 12000],
    [32, 8, 25000],
    [21, 3, 8000],
    [40, 12, 45000],
    [28, 6, 18000],
    [35, 10, 32000]
])
```

Columns: Age | Number of Orders | Total Spending

Questions / Tasks

1. How will you calculate the average customer age?
2. How will you calculate the average number of orders?
3. How will you calculate the average customer spending?
4. How will you find the highest-spending customer?
5. How will you find the lowest-spending customer?
6. How will you find customers spending more than ₹20,000?
7. How will you find customers who have placed more than 5 orders?
8. How will you find the customer with the highest number of orders?
9. How will you calculate the standard deviation of spending?
10. How will you classify customers as Premium, Regular or Basic based on spending?
11. How will you convert the data into a Pandas DataFrame?
12. How will you sort customers by spending?
13. How will you find the top three customers by spending?

Project 9: IPL Player Performance Analysis

Analyze the performance of five cricket players.

Given Dataset

```
players = np.array([
    [450, 15, 12],
    [520, 18, 15],
    [300, 10, 8],
    [650, 20, 18],
    [400, 14, 10]
])
```

Columns: Runs | Matches | Wickets

Questions / Tasks

1. How will you calculate the average runs scored?
2. How will you find the highest run scorer?
3. How will you find the lowest run scorer?
4. How will you find the highest wicket taker?
5. How will you calculate runs per match for each player?
6. How will you find players with more than 400 runs?
7. How will you find players with more than 10 wickets?
8. How will you calculate the standard deviation of runs?
9. How will you classify players as Excellent, Good or Average based on runs?
10. How will you convert the results into a Pandas DataFrame?

Project 10: Hospital Patient Analysis

Analyze basic numerical patient information and vital signs. Use only the provided numerical data for analysis.

Given Dataset

```
patients = np.array([
    [25, 120, 80, 72],
    [45, 150, 95, 88],
    [32, 130, 85, 76],
    [60, 170, 110, 92],
    [50, 155, 100, 90]
])
```

Columns: Age | Systolic BP | Diastolic BP | Heart Rate

Questions / Tasks

1. How will you calculate the average age?
2. How will you calculate the average systolic and diastolic blood pressure?
3. How will you find the highest systolic blood pressure?
4. How will you find the lowest systolic blood pressure?
5. How will you find patients with systolic BP greater than 140?
6. How will you find patients with heart rate greater than 85?
7. How will you calculate the standard deviation of each numerical column?
8. How will you classify patients as High or Normal based on systolic BP?
9. How will you convert the data into a Pandas DataFrame?

Project 11: Movie Rating Analysis

Analyze movie ratings and number of votes.

Given Dataset

```
ratings = np.array([
    [8.5, 120000],
    [7.8, 95000],
    [9.1, 180000],
    [6.9, 70000],
    [8.2, 110000]
])
```

Columns: Rating | Number of Votes

Questions / Tasks

1. How will you calculate the average movie rating?
2. How will you find the highest-rated movie?
3. How will you find the lowest-rated movie?
4. How will you find movies with rating greater than 8?
5. How will you find movies with more than 100,000 votes?
6. How will you find the movie with the maximum number of votes?
7. How will you calculate the standard deviation of ratings?
8. How will you classify movies as Excellent, Good or Average based on rating?
9. How will you convert the data into a Pandas DataFrame?
10. How will you sort the DataFrame by rating?